Prime Divisors of Reverse Concatenations

An Elementary Proof of the Conjecture in OEIS A077183

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Abstract

Let $R(k)$ be the integer obtained by concatenating the decimal representations of $k, k-1, \ldots, 1$, in that order. We prove that every prime other than 2 and 5 divides $R(k)$ for infinitely many positive integers k. This establishes the conjecture recorded in OEIS A077183 that the least such k exists for every prime $p \geq 7$. The proof uses Fermat's little theorem and finite geometric sums: within a sufficiently long block of integers of one digit length, a selected set of reverse concatenations attains every residue modulo p.

Keywords: prime divisors; reverse concatenation; Fermat's little theorem; finite geometric sums; A077183.
OEIS entry: A077183.

1 Definitions and the OEIS conjecture

For $m \geq 1$, let $\ell(m) = \lfloor \log_{10} m \rfloor + 1$ be its decimal digit length. Set $L(0) = 0$ and $L(k) = \sum_{m=1}^{k} \ell(m)$. Then reverse concatenation is expressed exactly by

$$R(0) = 0, \qquad R(k) = \sum_{m=1}^{k} m\, 10^{L(m-1)} \qquad (k \geq 1). \tag{1}$$

For example, $R(2) = 21$, $R(5) = 54321$, and $R(11) = 1110987654321$. Write p_n for the nth prime. OEIS A077183 defines $a(n)$ as the least positive k such that $p_n \mid R(k)$, or 0 if no such k exists. The entry attributes to Ryan Propper (29 July 2005) the conjecture $a(n) > 0$ for every $n > 3$ [1]. Equivalently, every prime $p \geq 7$ divides at least one reverse concatenation. We prove the following stronger statement.

2 Claim

Theorem 1. *For a prime p, there are infinitely many positive integers k with $p \mid R(k)$ if and only if $p \notin \{2, 5\}$.*

3 A finite geometric-sum identity

Lemma 1. *Suppose $b \in \mathbb{F}_p$, $b \neq 1$, and $b^t = 1$ for a positive integer t. Then, in $\mathbb{F}_p$,*

$$\sum_{i=0}^{t-1} b^i = 0, \qquad \sum_{i=0}^{t-1} i b^i = \frac{t}{b-1}. \tag{2}$$

Division denotes multiplication by the inverse of $b-1$ in $\mathbb{F}_p$.

Proof. The first identity follows by multiplying the sum by $b-1$. For the second, write $S = \sum_{i=0}^{t-1} i b^i$. Shifting the index gives

$$(b-1)S = (t-1)b^t - \sum_{i=1}^{t-1} b^i = (t-1) - (-1) = t.$$

Since $b-1$ is nonzero, it is invertible, proving the result. □

4 Proof of Theorem 1

Proof. Every $R(k)$ with $k \geq 1$ ends in the digit 1, so neither 2 nor 5 divides it. Modulo 3, equation (1) and $10 \equiv 1$ give $R(k) \equiv k(k+1)/2$. Thus all positive $k \equiv 0$ or $2 \pmod 3$ work.

Now fix a prime $p \geq 7$ and put $t = p - 1$. Choose $d \geq 2$ satisfying

$$d \equiv 1 \pmod t, \qquad 9 \cdot 10^{d-1} \geq pt. \tag{3}$$

There are arbitrarily large such d. Set $A = 10^{d-1}$ and let b be the residue of 10^d in $\mathbb{F}_p$. Fermat's little theorem gives $b = 10$ in $\mathbb{F}_p$, so $b \neq 0, 1$ because $p \geq 7$. It also gives $b^t = 1$. The integers $A, A+1, \ldots, A+pt-1$ all have exactly d decimal digits: the last is at most $10^d - 1$ by (3). Let $u = 10^{L(A-1)}$ in $\mathbb{F}_p$, which is nonzero. Equation (1) therefore implies, for $1 \leq j \leq p$,

$$R(A + jt - 1) \equiv R(A-1) + u \sum_{i=0}^{jt-1} (A+i) b^i \pmod p. \tag{4}$$

Indeed, each newly prepended integer has d digits, so its place value is multiplied by 10^d at each step.

Split the sum in (4) into j groups of length t. Since $b^t = 1$, Lemma 1 gives, in $\mathbb{F}_p$,

$$\begin{aligned} \sum_{i=0}^{jt-1} (A+i) b^i &= \sum_{h=0}^{j-1} b^{ht} \left((A+ht) \sum_{r=0}^{t-1} b^r + \sum_{r=0}^{t-1} r b^r \right) \\ &= \frac{jt}{b-1}. \end{aligned}$$

Consequently,

$$R(A + jt - 1) \equiv R(A-1) + jc \pmod p, \qquad c = \frac{ut}{b-1} \in \mathbb{F}_p^{\times}. \tag{5}$$

Here $c \neq 0$ because $u \neq 0$, $t = p - 1 \neq 0$ in $\mathbb{F}_p$, and $b - 1 \neq 0$. As j runs from 1 to p, the right-hand side of (5) attains every residue modulo p exactly once. In particular, one is zero. The resulting $k = A + jt - 1$ satisfies $k \geq 10^{d-1}$. Since admissible d are arbitrarily large, the resulting k are unbounded, proving infinitude. □

Corollary 1. *The OEIS conjecture $a(n) > 0$ for every $n > 3$ holds. In fact, the only zero terms of A077183 are $a(1) = a(3) = 0$, and $a(2) = 2$.*

Justification. The first three primes are $2, 3, 5$; $R(1) = 1$ and $R(2) = 21$. The remaining assertions follow from the theorem.

5 Example and attribution

For $p = 11$, the OEIS entry lists the least index as $k = 14$. The divisibility can be checked directly:

$$R(14) = 1413121110987654321 = 11 \cdot 128465555544332211.$$

The cited OEIS entry described the existence statement as a conjecture when consulted. This note establishes that statement and its infinitude strengthening; no claim of publication priority is made.

AI authorship and verification disclosure

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References

[1] The OEIS Foundation Inc., *Sequence A077183*. https://oeis.org/A077183. Conjecture credited to Ryan Propper, 29 July 2005. Accessed 3 October 2026.